CATATAN

- Net berlabel sama = terhubung (mis. +3V3, VBAT+).
- Semua simbol ground = common ground (0 V).
- NRF24 & MPU6050 disuplai 3V3 dari ESP32.
- ESC & motor disuplai LiPo (VBAT, Deans/T).
- ESP32 disuplai +5V dari buck converter.
- ESC: SIG↔IO25/26, GND↔GND.

⚠ +5V servo-lead ESC JANGAN ke ESP32.

⚠ NRF CE/CSN di GPIO4/5 (hindari GPIO6–11=flash).

LEGENDA

- SPI → NRF24
- I2C → MPU6050
- PWM → ESC
- +3V3 / +5V
- VBAT (baterai)
- Ground (GND) — = GND

Baterai LiPo
3S / 11.1V
(Deans / T-plug)
VBAT+
VBAT–

Buck Converter
VBAT → 5V output
VBAT+ in
+5V

ESP32-WROOM-32
5V / VIN
GND
3V3
IO23 · MOSI
IO19 · MISO
IO18 · SCK
IO5 · CSN
IO4 · CE
IO21 · SDA
IO22 · SCL
IO25 · ESC-L
IO26 · ESC-R
U1 (Microcontroller)

NRF24L01+
VCC
CE
CSN
SCK
MOSI
MISO
IRQ
GND
U2 (Radio 2.4GHz)

MPU6050
VCC
GND
SCL
SDA
U3 (IMU)

ESC 30A · KIRI (IO25)
SIG
GND
VBAT
M1
Motor BLDC

ESC 30A · KANAN (IO26)
SIG
GND
VBAT
M2
Motor BLDC

Robot Akuatik Feeder — Wiring	REV: 2.1
Drawn By: Samuel Alfonsus	Sheet: 1/1
PENGEMBANGAN ROBOT AKUATIK	2026-06-19